Project Report: CineIQ

An Explainable & Sentiment-Aware Movie Recommendation Engine

1. Introduction & Problem Statement

Streaming services today rely heavily on engagement metrics that often optimize for screen time rather than user satisfaction. This creates a paradox of choice where users spend significant time scrolling without finding relevant content. Furthermore, the rationale behind *why* a movie is recommended is rarely disclosed.

Objectives of CineIQ:

- **Break the Loop:** Utilize a hybrid ensemble to introduce serendipity alongside established preferences.
- **Gauge True Reception:** Re-rank recommendations based on actual audience sentiment using advanced NLP, rather than relying solely on box office numbers or static ratings.
- **Provide Transparency:** Implement an explainability layer so users understand the exact thematic or historical overlaps driving their recommendations.

2. Data Acquisition & Preprocessing

To build a robust and diverse recommendation engine, CineIQ leverages three distinct, large-scale datasets:

2.1 Datasets Utilized

- **MovieLens 25M:** The foundational dataset for user-item interactions, providing millions of historical movie ratings.
- **TMDB 5000 Metadata (Kaggle):** Provides rich metadata including cast, crew, genres, keywords, and plot overviews.
- **IMDB 50K Movie Reviews (Kaggle):** A dataset of binary-labeled reviews used to train and evaluate the sentiment analysis models.

2.2 Preprocessing & Filtering

To ensure computational efficiency and reduce noise from "long-tail" anomalies, the data underwent rigorous cleaning:

- **Interaction Thresholds:** The MovieLens dataset was filtered to include only users and movies with a minimum of 20 ratings. This reduced the dataset to a highly dense matrix of **142,274 users** and **16,744 movies**.
- **Metadata "Soup" Creation:** For the content-based engine, a unified text string (the "soup") was generated for each movie by combining lowercased, space-stripped genres, keywords, top 5 cast members, and the director (weighted heavily by repeating it three times).

3. System Architecture & Methodology

CineIQ uses a weighted ensemble architecture, passing candidates through multiple scoring layers before serving the final output.

3.1 Content-Based Filtering

This layer ensures thematic relevance to a user's seed query.

- **Vectorization:** A TF-IDF (Term Frequency-Inverse Document Frequency) Vectorizer was fitted on the metadata "soup," capped at 15,000 features using 1-to-2 n-grams.
- **Similarity Metric:** Cosine similarity was computed across the resulting matrix to find nearest neighbors.
- **Performance:** A sanity check evaluating Precision@10 for genre overlap yielded strong results, with movies like *The Dark Knight* achieving a perfect 1.0 score and a mean precision of 0.70 across test samples.

3.2 Collaborative Filtering

To capture latent user tastes that go beyond explicit metadata, CineIQ implements Matrix Factorization.

- **Model:** TruncatedSVD (Singular Value Decomposition) from scikit-learn.
- **Dimensionality:** Reduced the massive user-item matrix to n_components=100, capturing approximately 20.32% of the variance.
- **Evaluation:** On a 20% held-out test set, the SVD model achieved a Root Mean Square Error (RMSE) of **2.8590** and a Mean Absolute Error (MAE) of **2.4988**.

3.3 Sentiment-Aware Re-Ranking

A high metadata match does not guarantee a high-quality film. CineIQ adjusts scores based on audience sentiment.

- **Baseline Model:** VADER Sentiment Analysis was initially tested, achieving an accuracy of **67.32%**.
- **Deep Learning NLP:** To improve contextual understanding, a HuggingFace **DistilBERT** sequence classification model was fine-tuned on the IMDB dataset (42,500 training samples, 7,500 validation samples).
- **Training Details:** Trained over 3 epochs with a batch size of 32 and a learning rate of 2e-5.
- **Result:** The DistilBERT model significantly outperformed the baseline, achieving a final evaluation accuracy of **88.68%**.

3.4 Hybrid Ensemble Scoring

The final recommendation score is a weighted synthesis of the aforementioned engines:

1. Content score and SVD predictions are normalized to a 0-1 scale.
2. An ensemble score is calculated: $(0.5 \times \text{Content}) + (0.5 \times \text{SVD})$.

3. The final score adjusts the ensemble score using the DistilBERT sentiment normalization (default weight of 0.15).

4. Explainable AI (XAI) Layer

A core pillar of CineIQ is transparency. The application surfaces human-readable justifications for every recommendation.

4.1 Rule-Based Explanations

For straightforward matches, the system compares the metadata of the seed movie and the recommended movie to generate conversational explanations.

> *Example:* "Recommended because it is directed by the same filmmaker (Christopher Nolan), and is also a Crime and Action film, and has 72% content similarity."

4.2 LIME (Local Interpretable Model-agnostic Explanations)

For more complex relationships, CineIQ utilizes LIME to peek inside the "black box."

- A surrogate Logistic Regression model was trained on the TF-IDF features to predict high vs. low ratings (accuracy: 71.7%).
- LIME perturbates the input text to isolate the specific words driving the recommendation (e.g., isolating "tomhardy" or "thief" as key signals for the movie *Inception*).

5. Deployment & User Interface

The underlying logic and models are heavily abstracted from the user to provide a seamless experience.

- **Backend:** A robust **FastAPI** application serves the model endpoints (e.g., `/recommend` and `/similar`), handling the matrix lookups and DistilBERT inference efficiently.
- **Frontend Dashboard:** Built with **Streamlit** and **Plotly**, the interactive UI allows users to input their preferences and explore the XAI outputs.
- **Taste Visualization:** The dashboard goes beyond simple lists, offering visualizations of user behavior, including radar charts for genre affinities and historical breakdown by cinematic decades.
- **Experiment Tracking: MLflow** was utilized throughout the development lifecycle to track hyperparameters, SVD components, and variance metrics.

6. Conclusion

CineIQ successfully demonstrates the viability of an open, multi-layered recommendation system. By moving away from closed-loop collaborative filtering and injecting real-world NLP sentiment analysis, the engine produces high-quality, relevant results. More importantly, the integration of rule-based templates and LIME establishes a paradigm of trust; users are not just told *what* to watch, but they are empowered to understand *why* it was suggested to them.